

Applications: Terminal - root@codesp... Tue 2 Jun, 17:16 root

```
root@codespaces-fd3384:~/Desktop# vim unsignedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -march=rv64i -o unsignedHighest.o unsignedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk unsignedHighest.o
bbl loader
highest number represented by unsigned long long int is 18446744073709551615
root@codespaces-fd3384:~/Desktop# vim unsignedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -march=rv64i -o unsignedHighest.o unsignedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk unsignedHighest.o
bbl loader
highest number represented by unsigned long long int is 18446744073709551615
root@codespaces-fd3384:~/Desktop# vim unsignedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -march=rv64i -o unsignedHighest.o unsignedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk unsignedHighest.o
bbl loader
highest number represented by unsigned long long int is 1023
root@codespaces-fd3384:~/Desktop# vim unsignedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -march=rv64i -o unsignedHighest.o unsignedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk unsignedHighest.o
bbl loader
highest number represented by unsigned long long int is 0
root@codespaces-fd3384:~/Desktop# vim unsignedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -march=rv64i -o unsignedHighest.o unsignedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk unsignedHighest.o
bbl loader
highest number represented by long long int is -1024
root@codespaces-fd3384:~/Desktop# vim signedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -march=rv64i -o signedHighest.o signedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk signedHighest.o
```

29°C Mostly cloudy 22:46 02-06-2026

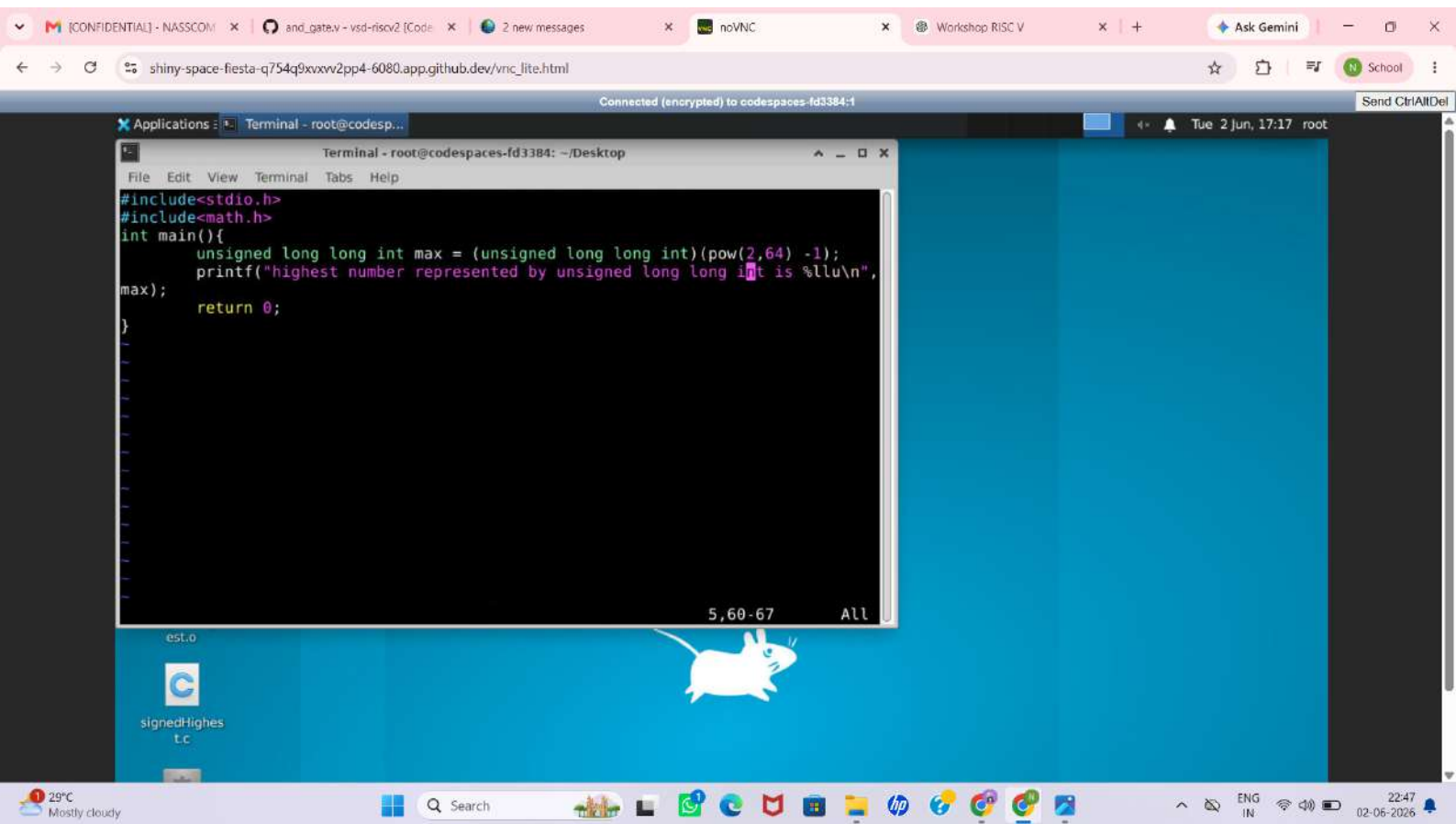
Applications: Terminal - root@codesp... Tue 2 Jun, 17:16 root

Terminal - root@codespaces-fd3384: ~/Desktop

```
root@codespaces-fd3384:~/Desktop# spike pk unsignedHighest.o
bbl loader
highest number represented by unsigned long long int is 0
root@codespaces-fd3384:~/Desktop# vim unsignedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -mar
ch=rv64i -o unsignedHighest.o unsignedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk unsignedHighest.o
bbl loader
highest number represented by long long int is -1024
root@codespaces-fd3384:~/Desktop# vim signedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -mar
ch=rv64i -o signedHighest.o signedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk signedHighest.o
bbl loader
highest number represented by long long int is 2147483647
lowest number represented by long long int is -2147483648
root@codespaces-fd3384:~/Desktop# vim signedHighest.c
root@codespaces-fd3384:~/Desktop# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -mar
ch=rv64i -o signedHighest.o signedHighest.c
root@codespaces-fd3384:~/Desktop# spike pk signedHighest.o
bbl loader
highest number represented by long long int is 9223372036854775807
lowest number represented by long long int is -9223372036854775808
root@codespaces-fd3384:~/Desktop#
```

esto
signedHighes
t.c

29°C Mostly cloudy Search 22:46 02-06-2026



Browser tabs: [CONFIDENTIAL] - NASSCOM, and_gata.v - vsd-riscv2 [Code], 2 new messages, noVNC, Workshop RISC V, Ask Gemini, School.

Address bar: shiny-space-fiesta-q754q9xvww2pp4-6080.app.github.dev/vnc_lite.html

Terminal window: root@codespaces-fd3384: ~/Desktop

```
#include <stdio.h>
#include <math.h>
int main() {
    long long int max = (long long int) (pow(2,63) -1);
    long long int min = (long long int) (pow(2,63) * -1);
    printf("highest number represented by long long int is %lld\n", max);
    printf("lowest number represented by long long int is %lld\n", min);
    return 0;
}
```

Terminal status: "signedHighest.c" 10L, 313B 5,32-39 All

Desktop icons: esto, signedHighes, t.c

Taskbar: 29°C Mostly cloudy, Search, Windows icons, ENG IN, 22:47 02-06-2026

Applications: Xfce Terminal

Terminal - root@codespaces-fd3384: /workspaces/vsd-riscv2/samples

```
File Edit View Terminal Tabs Help
root@codespaces-fd3384:~/Desktop# cd /workspaces/vsd-riscv2/samples
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# cat sumlton.c
#include <stdio.h>

int main(){
    int i, sum=0, n=20;
    for(i=1;i<=n;i++)
        sum = sum + i;
    printf("Sum from 1 to %d is %d \n",n,sum);
    return 0;
}

root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -O1 -mabi=lp64 -march=rv64i -o sumlton.o sumlton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# ls -ltr sumlton.o
-rwxrwxrwx 1 root root 167512 Jun 1 15:33 sumlton.o
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -Ofast -mabi=lp64 -march=rv64i -o sumlton.o sumlton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples#
```

Mon 1 Jun, 16:04 root

Send Ctrl+Alt+Del

100dc: ffff0797 auipc a5,0xffff0
100e0: f2478793 addi a5,a5,-220 # 0 <main-0x100b0>

29°C Partly cloudy

21:34 01-06-2026

Applications 2 Xfce Terminal

```
Terminal - root@codespaces-fd3384: /workspaces/vsd-riscv2/samples
File Edit View Terminal Tabs Help

root@codespaces-fd3384:~/Desktop# cd /workspaces/vsd-riscv2/samples
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# cat sum1ton.c
#include <stdio.h>

int main(){

    int i, sum=0, n=20;
    for(i=1;i<=n;i++)
        sum = sum + i;
    printf("Sum from 1 to %d is %d \n",n,sum);
    return 0;
}

root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -
01 -mabi=lp64 -march=rv64i -o sum1ton.o sum1ton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# ls -ltr sum1ton.o
-rwxrwxrwx 1 root root 167512 Jun  1 15:33 sum1ton.o
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -
0fast -mabi=lp64 -march=rv64i -o sum1ton.o sum1ton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples#
```


Applications ▢ 2 Xfce Terminal

```
Terminal - root@codespaces-fd3384: /workspaces/vsd-riscv2/samples
File Edit View Terminal Tabs Help

root@codespaces-fd3384:~/Desktop# cd /workspaces/vsd-riscv2/samples
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# cat sum1ton.c
#include <stdio.h>

int main(){

    int i, sum=0, n=20;
    for(i=1;i<=n;i++)
        sum = sum + i;
    printf("Sum from 1 to %d is %d \n",n,sum);
    return 0;

}

root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -
01 -mabi=lp64 -march=rv64i -o sum1ton.o sum1ton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# ls -ltr sum1ton.o
-rwxrwxrwx 1 root root 167512 Jun  1 15:33 sum1ton.o
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -
0fast -mabi=lp64 -march=rv64i -o sum1ton.o sum1ton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples#
```

```
Terminal - root@codespaces-fd3384: /workspaces/vsd-riscv2/samples
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sumlton.o:      file format elf64-littleriscv

Disassembly of section .text:

000000000000100b0 <main>:
 100b0:      00021537      lui      a0,0x21
 100b4:      ff010113      addi     sp,sp,-16
 100b8:      0d200613      li       a2,210
 100bc:      01400593      li       a1,20
 100c0:      18050513      addi     a0,a0,384 # 21180 <__clzdi2+0x48>
>
 100c4:      00113423      sd       ra,8(sp)
 100c8:      340000ef      jal      ra,10408 <printf>
 100cc:      00813083      ld       ra,8(sp)
 100d0:      00000513      li       a0,0
 100d4:      01010113      addi     sp,sp,16
 100d8:      00008067      ret

000000000000100dc <register_fini>:
 100dc:      ffff0797      auipc    a5,0xffff0
 100e0:      f2478793      addi     a5,a5,-220 # 0 <main-0x100b0>
:
```



```
Terminal - root@codespaces-fd3384: /workspaces/vsd-riscv2/samples
File Edit View Terminal Tabs Help

root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -
Ofast -mabi=lp64 -march=rv64i -o sumlton.o sumlton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# gcc sumlton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# ./a.out
Sum from 1 to 20 is 210
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -
Ofast -mabi=lp64 -march=rv64i -o sumlton.o sumton.c
riscv64-unknown-elf-gcc: error: sumton.c: No such file or directory
riscv64-unknown-elf-gcc: fatal error: no input files
compilation terminated.
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -
Ofast -mabi=lp64 -march=rv649 -o sumlton.o sumlton.c
cc1: error: -march=rv649: first ISA subset must be `e', `i' or `g'
cc1: error: ABI requires -march=rv64
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# riscv64-unknown-elf-gcc -
Ofast -mabi=lp64 -march=rv64i -o sumlton.o sumlton.c
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# spike pk sumlton.o
bbl loader
Sum from 1 to 20 is 210
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# spike -d pk sumlton.o
(spike) until pc 0 100b0
bbl loader
(spike) reg 0 a2
0x0000000000000000
```

```
Terminal - root@codespaces-fd3384: /workspaces/vsd-riscv2/samples
File Edit View Terminal Tabs Help
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# spike -d pk sumlton.o
(spike) until pc 0 100b0
bbl loader
(spike) reg 0 a2
0x0000000000000000
(spike)
core 0: 0x000000000000100b0 (0x00021537) lui      a0, 0x21
(spike) reg 0 a0
0x000000000000021000
(spike)
core 0: 0x000000000000100b4 (0xff010113) addi    sp, sp, -16
(spike) reg 0 sp
0x000000007f7e9b40
(spike) q
root@codespaces-fd3384:/workspaces/vsd-riscv2/samples# spike -d pk sumlton.o
(spike) until pc 0 100b8
bbl loader
(spike) reg 0 sp
0x000000007f7e9b40
(spike)
core 0: 0x000000000000100b8 (0x0d200613) li      a2, 210
(spike) reg 0 sp
0x000000007f7e9b40
(spike)
```